2.5 Graphical representation of motion along a straight line

Plotting the distance/displacement or speed/velocity on a graph helps us to understand certain things about time and position.

2.5.1 The distance – time graph for Uniform motion

Consider the Table 2.1 which shows the distance walked by a person at different times.

Table 2.1 Uniform motion

Time (minute)	Distance (metre)
0	0
5	500
10	1000
15	1500
20	2000
25	2500

A graph is drawn by taking time along X-axis and distance along Y-axis. The graph is known as distance – time graph.

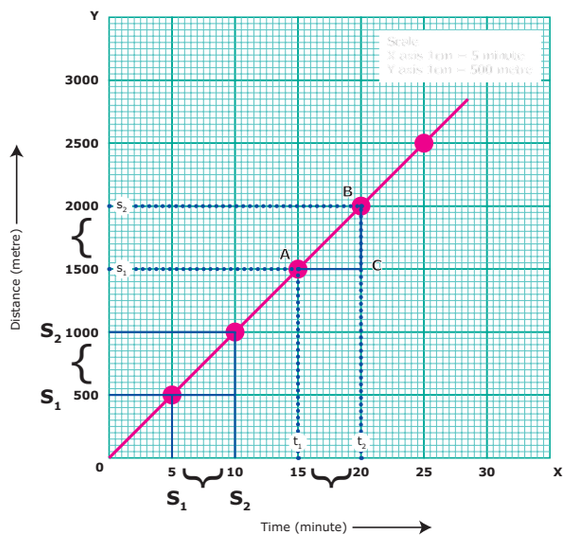


Figure 2.2 The distance – time graph for Uniform motion

When we look at the distance – time graph, we notice few things. First, it is a straight line. We also notice that the person covers equal distances in equal intervals of time. We can therefore conclude that he walked at a constant speed. Can you find the speed at which he walked, from the graph? Yes, you can. The parameter is referred as the slope of the line.

Speed = Distance covered / Time taken
= Slope of the straight line
= BC/AC (From the graph)
= 500 / 5 = 100 m/min

Steeper the slope (in other words the larger value) the greater is the speed.

Let us take a look at the distance – time graphs of three different people – Asher walking, Saphira cycling and Kanishka going in a car, along the same path (Fig 2.3). We know that cycling can be faster than walking and a car can go faster than a cycle. The distance – time graph of the three would be as given in the following graph. The slope of the line on the distance – time graph becomes steeper and steeper as the speed increases.

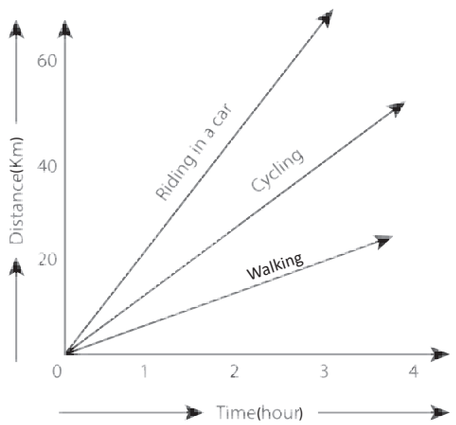


Figure 2.3 Comparison of speed

2.5.2 The distance time graph for Non-uniform motion

We can also plot the distance – time graph for accelerated motion (non-uniform motion). Table 2.2 shows the distance travelled by a car in a time interval of two seconds.

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